

# Scripting for Offense & Defense

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## The Analyst's Daily Loop

**Collect** – logs, packets, endpoints, alerts

**Normalize** – put it in one format

**Enrich** – add context (geo, whois, blocklists)

**Analyze** – patterns, anomalies

**Act/Roll back** – block, alert, escalate

Everything above is scripts.

## Practical Collection Script

```
import datetime, json

logs = []

with open("/var/log/auth.log") as f:
    for line in f:
        if "Failed password" in line:
            parts = line.split()
            logs.append({
                "time": parts[0] + " " + parts[1],
                "user": parts[-3] if len(parts) > 3 else "?",
                "ip": parts[-4] if len(parts) > 4 else "?",
                "source": line.strip(),
            })

print(json.dumps(logs[:3], indent=2))
print("total:", len(logs))
```

## Enrichment: WHOIS / DNS in Python

```
import socket

try:
    ip = socket.gethostbyname("evil.example.com")
    print("resolved:", ip)
```

```
except socket.gaierror:  
    print("no resolution")
```

Real analysts query VirusTotal, Shodan, WHOIS APIs – but always for authorized purposes and with API keys stored safely.

## Defense Automations You Will Write

**Baseline diffing** – snapshot listening ports, files, processes daily; alert on diffs

**Alert on new admin users:** parse `Get-LocalUser` (PowerShell) and compare Hash a directory daily to detect file modification

**Parse SIEM output** into a Python dashboard

## Offense Automation You Will Write (Lab-Only)

**Recon loop:** iterate subdomains, scan each discovered host

**Password spray** (careful, CREDENTIAL-STUFFING) against your OWN test app with throttling

**Nmap parser:** run nmap, parse the XML output (`-oX`), summarize services

**Exploit helper:** pass a banner → searchsploit output → candidate CVE list

## The Automation Golden Rules

**Never automate against systems you lack permission for** – scripted attacks hit many targets fast and are LOUD.

**Always throttle & log** – a fast scanner = DoS + detection.

**Idempotent + rerunnable** – re-run scripts safely.

**Determinism** – same input → same output; no surprises.

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[← SQL Basics: Reading & Attack...](#)

Module complete ✓